Unit 5 - Managing Networks
Assignment 12: Managing a LAN Tasks 3&4

Introduction

Task description - *You are a junior consultant for T5 Alliance. The work you have done so far for Competition Bikes Ltd has been very well received. As they have no IT support staff of their own, they are interested in outsourcing the ongoing management to your company. First, however, you must prove your knowledge of network management.*

Task 3

(U5M2) - Outline the purpose of networking tools

Task description - *For this task you need to outline the goals of fault management (i.e. what should fault management actually achieve). Explain why it is important and what could happen if it wasn't done properly/regularly.*

- *Identify the goals of fault management.*
- *State why fault management is important.*
- *Describe what could happen if fault management wasn't done.*

Introduction

Fault management is the use of a passive system within a network that constantly monitors the performance of the hardware and software found within a network, determining whether they are performing above or below the standard performance rate.

Basically, a fault management system monitors the network for any fluctuations within the network, either fixing said error or warning the technician/administrator that there is a performance problem, containing brief information of the cause and type of error.

Reacting accordingly either by maintaining performance autonomously or assisting the administrator, with an error message, on the errors found during the hours that it's being operated on.

This is so it averts the hardware from failing before further damages or errors, which could render the system inadequate for use, by indicating to the administrator that there is a problem via error message.

Identify the goals of fault management.

In terms of fault management, there are several aims/goals that are vital to keep a network functioning at an optimal standard. However there are key main aims of fault management that are vital for maintaining a stable network, these aims can range from: Monitoring and reporting the status of a network component to the administrator; The use of alarms to alert the administrator of problems (such as a decrease in network performance); Logging faults and repairs; Remotely controlling network components so administrators can reboot, repair or shut down problematic components.

The first aim mentioned, which is the monitoring and reports of the status of a network component to the administrator. This allows the administrator to see how the network is performing. The way that this is done depends on the type of network tool that is currently being used, with some being more user friendly in monitoring than others.

However, while one type of software tool operates different from others, they all have the same goal. Which is to monitor and report the status, that could be accomplished with the fault management system that provides regular reports on network components. Including current statistics, performance of the component and other previous reports.

The second aim mentioned is the use of alarms to alert the administrator of problems or decrease in network performance. Which is done by the software with different types of ways of warning the administrator, depending of the type of software, the most common being a pop up alert box.

Basically, the fault management systems uses alarms to notify the administrator of a fault in the network, allowing the administrator to fix the fault and keep the network running at optimal level. Also, depending of the software it would send alerts to any type of error (if the software isn't programmed to deal with certain errors), or it would send alerts if it does not have the capability to deal with an error (if it is programmed to deal with errors).

The next aim mentioned is the ability to log faults and repairs. Essentially, this is the recording of errors and what was done to said error in the software. This gives the ability to be revised later on to check what caused the error, what was done to fix the said error and whether it was the most efficient method.

So, if a fault appears in the log multiple times, it will allow for the fault to be fixed faster and/or shows that the component with the fault may need replacing. This would allow the system to be restored quickly with minimal downtime, which is a vital aim of fault management.

The final aim mentioned is to remotely control network components, which is the ability for administrators can reboot, repair or shut down problematic components. Essentially, the admins have the ability to take control of a specific component and fix the error with any means necessary, even if it means turning the component off.

These aims are all in place to reach one main goal, which is to protect the information and integrity of data for a company. This is so they are able to keep the data from being corrupted, deleted or stolen. Which is vital because a company would need essential data in order to make a profit and survive.

State why fault management is important

A successful fault management system is vital for a network to function at optimum level, since there can be severe consequences if the fault management system isn't implemented properly. Basically, fault management is important because it has the ability to detect and resolve problems allows the network to be maintained to a high standard and fixed quickly and efficiently if a problem occurs, which can save companies valuable time and money.

For example, the first goal mentioned previously "monitoring and reporting errors", allows the administrator to assess the network's performance. Determining any serious fluctuations which could cause critical negative impacts for a network and its integrity. Furthermore, having this goal reduces the chance of an error by predicting a fault before occuring and it keeps the network a lot more secure.

In terms of maintaining a network, with the second goal mentioned, alerts are arguably one of the most important goal for the fault management system. The reason for this is that it alerts the administrator that there is a problem in the first place. Without this the admin would be clueless about the type of errors that are currently active in the network or whether there is an error.

With the third goal mentioned, having reports of faults and repairs allows the administrator to find out about what error that is currently occuring and determining the fastest solution. However, this is available only if a problem has occurred and has been reported once or multiple times. Although, if the admin does find an error and removes the error correctly, the logs could be used to help in the future for other administrators.

With the final goal mentioned, having the ability to remotely control network components allows the administrator to deal with the said component quickly, reducing the effect the problem has on the network, also reducing downtime if the problem is critical.

In retrospect, all these goals are important for maintaining a network so revenue can be made within the business the network is found in, which is the true main aim of fault management. Even though the goals fault management is centered in keeping a network functioning, the main aim is to keep the company's network intact so it can survive in the modern business world.

Describe what could happen if fault management wasn't done

A network without any sort of fault management isn't more likely to fail than a network with it. However, the network with the fault management in place has the ability to solve those failures and keep the network running longer. Networks without fault management would pick up on this slower, meaning that solving the problem would be harder and could potentially damage the network.

To elaborate, with these errors appearing within the network, the network could become slower and would not be able to support the amount of users accessing the network, causing the network to potentially crash and even fatally damage the data and information within the network. Which would cause a catastrophic problem for the company, affecting the work of staff and the business's revenue being affected.

Another consequence for not resolving faults is that the security system might be affected, leaving the network open for external attacks by anonymous attackers. To prevent this backup software and hardware can be used.

In problems such as this, the company should have sufficient backup hardware to replace broken components. If there is not a backup component, time and money will be wasted by having to buy a new component and configure it to work with the network. Also, the company would need backup software in case of system failure or power failure.

Also, even if there is the implementation of fault management but the administrator does not have the knowledge effectively manage the faults in the network, the number of faults could also increase. The reason for this is due to the fact that fault management allows administrators to detect and repair weaknesses and threats in the network, because it gives the administrators the ability to keep the network running at optimum level.

All in all, for the company, the lack of fault management would be a costly and time consuming problem for the network and the vital data found in the network. The reason being that the company would have to spend a lot of time and money trying to resolve issues that will occur on the network.

Which can also waste time on technicians and network managers, making them unable to carry out other jobs in the meantime. Which could bring in more problems for the company outside of the network aspect, increasing the waste in costs and time.

Conclusion

In conclusion, Fault management is a vital part for the network's health and should be implemented for network management. Since it provides a range of techniques that keep the network working efficiently and it minimises the possibility of the network failing, which could cost the company valuable assets such as money, time and the workforce.

An example to prove this concept is if the network does fail, causing the business to lose access to the rest of the data found in the network, meaning that the valuable data cannot be used for profitable reasons. Also, the staff wouldn't be able to access the network in order to carry out their jobs, causing the company to lose valuable time to make profits in. If the network has the fault management system implemented, it helps reduce all these factors for a business.

Overall, fault management is an aspect of a network that uses software to detect, isolate and resolve problems in a network, enabling the network to perform as well as possible. If installed correctly, fault management software has the capability to keep the network running at an optimum level which benefits both the profits of the company and the employees.

Task 4

(U5D1) - Justify the inclusion of routine performance management activities within a network manager's role

Task description - *Explain why certain performance management activities should be performed. Explain why the following performance management activities should be performed.*

- *Performance monitoring*
- *Checking traffic*
- *Checking user response times*
- *Anti-virus scans*

Introduction

Routine performance management activities are essential and vital for a network, no matter the size or the usage of the network. This is so that the network administrator is always aware of the network's performance and that it is running efficiently, without the company being affected with profits and work time by the errors that the network could bring.

To elaborate, with the implementation of performance activities, it is important that the network administrator has to ensure all components of the network are functioning effectively and response time is as expected.

Meaning that the maintenance of performance gives the ability for facilities accessed by users, which should always be available and monitored in case a problem occurs. Giving users the ability to access their accounts, files, peripherals and the internet without their data and access to said data being interfered with. Hence why it is absolutely necessary for a business.

Performance monitoring

The whole purpose of performance monitoring is to monitor the usage and performance of a computer network, checking for slow or failing systems. To elaborate, if the performance of a section of the network is degrading, then the administrator may choose to reconfigure or replace devices to increase the performance of that section.

Usually, the way this is done is by the system notifying the network administrator of any performance issues or outages with some kind of an alarm or an email. Usually by the admin being able to spot out any early signs of performance issues or the tool will send regular reports to the administrator which would be currently monitoring the network.

For example, tools such as Microsoft Systems Centre or other tools from other companies such as Solar Winds, give the ability to diagnose the performance tasks. Sometimes the tools will include past reports to compare how each section is performing now to how it was performing the last time a report was sent, allowing the administrator to determine whether the network can be improved or not based on the sections performance.

While different tools have different methods, what they have in common is that the performance management system reports on different sections of a network and how they are performing. Usually, as mentioned previously, it will include past reports to compare how each section is performing to the administrator's current time.

Basically, with this ability for comparison, the administrator and network manager would see how to improve it via software, hardware or if it can be improved at all. Indicating that improving performance may show that certain changes to the network have been beneficial and if applied to other sections of the network, overall network performance could be improved and make the network more efficient.

Overall, monitoring and then improving performance to the network is proven to be beneficial and, if applied to other sections of the network, overall network performance could be improved, making the network much more efficient.

Also, routinely checking the network helps spot or decrease the probability of errors in system performance. Missing problems such as decrease in network performance can affect the network quite severely.

Meaning that if a network component develops a problem and the performance of the component is not checked regularly, the problem might be missed, causing the device to break. If a problem happened to an important component such as a server, the whole network could be put under strain and might affect people using the network costing the organisation time and money.

Basically, there are consequences for not routinely checking network performance. Especially for a business point of view, in which this system has the potential of saving a lot of money and reduce many problems. It is one of the best possible ways to ensure that the business is operating properly, and without it would be catastrophic.

Checking traffic

There are several reasons that the traffic on the network should be checked and monitored at a constant rate. The reason being the information that is produced by certain network traffic monitoring tools could be used in multiple IT operational and even checking the network for any vulnerabilities.

Also, checking traffic can help troubleshoot network related issues, which gives the administrator the ability to analyze the impact that new applications, users or hardware can impact the overall network.

As mentioned before in other tools, not all the tools for monitoring network traffic are the same. Meaning that they are divided into two different types of tools. Which are deep packet inspection tools and flow-based tools. While these type of tools usually stores log data, they function differently on how they detect the network traffic.

While there are two different types of checking network traffic, some can still carry out the same purpose. Such as the identification of slow software, detecting spyware/hacking, detecting outflow of data from clients and even determine the bandwidth usage within the company.

To elaborate, detecting the outflow of data from clients and overall bandwidth usage is a key feature of checking the traffic. For instance, detecting outflow of information of users is applicable to the companies that deal with user data, which is a majority of companies.

A network with good security gives good data integrity, which never lets the data of the user to get leaked from its system or have an effect on bandwidth. Also, If the company is worried about the overall usage of the network and is unable to find out which department is using the data flow in significant amounts.

Certain network tools can use the system which detects the outflow of data from users, pointing out the IP addresses and sometimes the devices in which are using the network resources. In which the administrator would take appropriate action to reduce the pressure on the network bandwidth and data.

Therefore leading to the ability of checking the performance of applications, which is another main use of checking network traffic. Since performance plays an important role in the workers

ability to access and send information over a network, therefore playing a role in profits of the company.

For example, checking the traffic gives the administrator the ability to check whether an application is having slow performance or freezing/crashing. There could be multiple reasons which could cause the application to cause this issue, making the identification of the issue being both time consuming and costly.

However, there are applications that can filter and report the accurate cause of the application crashing, such as Netflow software. It does this by combining the internal data reports with the external resources, which the system administrator can learn about the system and the network suffering the error.

The way that spyware or other forms of hacking is detected, is by the administrators seeing a strange fluctuation in the data pattern. Which may sometimes be mistaken by a performance issue caused by a user downloading/uploading data.

With certain tools, especially those with some sort of data analyzer, these unusual patterns can be detected easily by the admin. However, if the networking tool does not contain a data analyzer, the patterns can sometimes go unchecked due to the fact that it can give a false sense of security to the administrator, mostly due to the administrator believing it is caused by a user within the network.

In a business point of view, these type of spyware usually won't cause any financial problems to the company, but it will give the company a bad image since their network has been infiltrated. However, in very rare cases, they could affect high rates of financial loss due to the data stolen and given to a competitor or deleted in general.

Overall, checking network traffic is vital for a network administrator who is not aware of the amount of broadcast traffic in the network. Without the ability to check the network traffic, the admin would not be able to take appropriate action in order to address possible errors.

For instance, if the broadcast traffic becomes excessive over time, either by users uploading or downloading too much data, the performance will reduce. This could be a gradual thing that could carry on consistently over multiple weeks and sometimes months. Meaning that checking traffic within the network is arguably as important as monitoring overall performance.

Checking user response times

Checking user response times is essentially the administrator using a network monitoring tool to check the real-time information on how users are interacting with the network. usually by applications they are using and the tool can even tell the admin where they are located.

The intention of this tool is to help managers and administrators help determine the link between the network performance and what impact it has on application performance. However, it is essential that the admin does not maintain a strict focus on the technology, otherwise there will be a lack of user friendliness with the application. Making applications “buggy” for users due to the large implementation of the monitoring tool.

Usually, there are certain helpful uses involved with checking the user response times that could benefit the network in certain aspects. These uses can range from the information obtained by response time, the quality of service, the data capacity used/throughput and the defects per interval.

The first and intentional use is the information that can be obtained by the user response times., this information can be used to help identify problems even before they occur. The problems can range from cloud storage, monthly software as a service (SaaS) problems and even applications found on the premises of the network.

The second use can be the quality of service, or QoS. This is a simple use that essentially gives the administrator the visibility into the delivered performance, however only if it is in relation to the expected performance which is based on the quality of service marking and enforcement.

The third potential use it can have is the ability to check the data capacity used or the throughput within the network and user’s PCs. This is done by the admin checking the application for a rough insight into the application-level problems, which can be checked to make sure when admins known when are where to apply additional resources.

Lastly, an important factor with network maintenance is the ability for the administrator to check the defects per interval. This is a very effective method in which the admin can use the tool to show when the errors, that are rooted in partial responses, can be traced. Errors such as content errors or missing components can be discovered through this method, which could have negative impact on network performance if it isn’t dealt with.

Overall, in simple terms, the user response times will be affected by the relative performance of the network. Therefore it gives the ability to determine multiple problems with this type of information from the user, such as quality of service, whether there are throughput/data capacity issues or any other forms of errors.

Without the ability to check for issues that this information could bring, it would lead to lost productivity as the end user has to deal with the unresponsive or “buggy” system to respond. Also, in much worse cases, the network has the possibility to stop working in general, therefore delaying the data report the issue, making it much more difficult for it to be resolved.

Essentially, networks that are not monitored for performance or checked for traffic will inevitable develop issues that will have a knock-on effect for the end user and eventually the entire company. Also, as mentioned earlier, work time is essential for company revenue and the lack of a workforce would have a growing significant impact on a company.

Especially since employees and even paying customers are most likely to be repelled to a system lacking in performance, therefore going for a competing company since they have better responsive systems.

Anti-virus scans

An essential part in keeping the network healthy, not only the integrity of the data and the network as a whole but also in optimal performing order, is the protection against viruses. Hence why Anti-virus scanning software is used in this area.

There are certain types of software tools that work by scanning through all of files stored with in a server, then checking if files contain any suspicious forms of data or looks suspicious by inspection. This tool can be a standalone anti-virus scanning software or it can be implemented into a large network management tool.

In simple terms, software such as this will have the ability to perform ‘quick scans’. Which will search a smaller but more specific part of the storage drives. For instance, files that are downloaded from the internet. Alternatively, a ‘full’ system scan will search every file located on each drive of the server found within the network.

To elaborate, anti-virus scans exist to insure that servers and equipment are continually up and running for the company’s customers and staff, meaning that regular virus scans are vital to insure that nothing has infected the systems via the network. Meaning that carrying out virus scans regularly insures that the data of the company is protected.

The reason why the administrator would scan for malicious software is due to the fact that lots of users on the network might download software without thinking about the risks involved or believing it is safe to download. Which can be mostly caused due to the name of the file. If this was the case and a miscellaneous program does infect the system, it could be that company’s details and files could in fact get stolen, removed or is rendered inaccessible due to corruption.

For example, a malware in the form of a worm, usually found with a standard app name with a .exe executable. Just by inspection doesn’t seem safe due to the file format, however could be easily disguised within other legitimate forms of .exe files. Meaning if the network was to have unattended malicious software, it may cause the network to reduce in performance significantly.

Inturn causing problems for the users on the network because they won't be able to carry out their daily tasks within the business, therefore causing a loss in profits for the business. Which the main reason that a network administrator would scan for malicious software.

However, there is also a risk that if network becomes infected with a virus, that users with sensitive information on the network might face a colossal risk. Due to the fact that they can be seen by unauthorised personnel, meaning that the employees can be traced for whatever type of information is left on the network.

Also the malicious software could lead to losing data that could be of vital importance for a company, especially if the company relies on client's information in order to carry out the business's intended purpose.

Overall, there are multiple performance and safety issues that can originate from malware that has the ability to negatively impact the performance of systems, render them unresponsive or any form of access is denied. Also, the colossal risks of the a significant amount of security issues they provide for the clients, employees and the company as a whole.

Essentially, performance monitoring and checking traffic may sometimes lead to malware as the culprit. However, it is essential that networks are automatically virus checked at very regular periods. So malware doesn't stay implemented into the network, causing damage to the data found in the network.

Apart from virus scans, this can usually be countered by the detection of malware through the monitoring of performance, giving the administrator the ability to check the traffic and whether it has a problem. The reason this can be done due to the fact the malicious code is already affecting the network since it was installed and executed, basically leaving some sort of epicentre.

Basically, having software with the ability of anti-virus scanning in place is a vitally important layer of performance and security protection that, if carried out appropriately, should ensure that malware does not enter the network. Making it essential to have installed in networks.

Conclusion

In conclusion, through research, it has been evident that various software tools are used routinely by administrators to perform certain tasks in order to keep the performance of the network in optimal working order. With most of the software tools being able to routinely observe the network, which allows the administrator to know exactly how the network performs.

For instance, the admin may spot when and where there are strange patterns, usually leading to a problem. This makes spotting and resolving problems easier for the administrator, since they know precisely what, where are why there is a problem currently occurring.

Also, by logging the performance of a network when it is functioning normally and when there is a change, means that they can compare performance records to help find errors. Meaning that routinely checking the network not only helps spot problems or decreases in system performance, but keeping a record of them also helps in the long term.

Essentially, logging performance is useful if a one device or a whole area of the network is upgraded, since it allows the administrator to compare the results and see if the new upgrade is beneficial. If it is proven successful, then other similar areas could be upgraded to improve the quality of the network.

However, there are consequences for not routinely checking network performance, such as loss of money. Since problems such as decrease in network performance can affect the network quite severely.

To elaborate, if a network component develops a problem and if the performance of the component is not checked regularly, the problem might be missed. Which can cause the device to be rendered dysfunctional, or could even escalate to the rest of the network depending on the type of problem found.

For instance, if a problem happened to an important component (such as a server), the entire network could be put under strain and might affect people using the network. Which could impact the company's profits because it will cost the organisation time and money.

Overall, network monitoring will save a company a significant amount of time, therefore translating to a significant amount of money. Without performance management, a lot of time would have to be spent investigating by the administrator and managers, which would result in more hours having to be worked. This will not only cost the company more money, but it will lower productivity.

With the ability to quickly point out and fix network issues, it is technically increasing the profits of the company. The reason being that everything is running smoother, giving the company more time to run the business.

Especially when then administrators understand how all of the devices are being used. They are able to identify what needs additional maintenance, so they can decrease the chance of an error occuring before the error actually occurring quickly and effectively.